

A Metric Hierarchy for Certifying Quantized Driving VLA Policies: Open-Loop, Coherence, and Route Competence All Mislead; Graded Closed-Loop Safety Decides

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Abstract—Vision–language–action (VLA) driving policies must be quantized before they can run on in-vehicle hardware, and quantization is almost always validated on open-loop accuracy. We show this validation is not merely optimistic but structurally blind—and that the natural intermediate proxies are blind too. Evaluating post-training weight quantization of OpenDriveVLA-0.5B with four metrics of increasing fidelity—open-loop L2, closed-loop generation coherence, closed-loop route competence on an adversary-free benchmark we construct, and graded closed-loop safety (impact-speed-graded NCAP score)—we find a strict hierarchy of blindness: each metric certifies quantized policies that the next rejects. Open-loop L2 certifies everything (a $19\times$ ego-ablation shows it is ego-extrapolation). Coherence certifies group-wise INT4, which restores well-formed trajectories but not safety. Route competence certifies AWQ-INT4, which drives routes at near-FP16 level (87% vs 94%) yet fails safety-critical scenarios exactly like naive INT4. On the graded safety metric, every 4-bit recipe we test—naive, group-wise, AWQ, and DCT-rotated W4A4—collapses to 2.6–2.9 of FP16’s 4.4 stars, with failures concentrated on long-horizon stationary-obstacle scenarios; per-channel INT8 is statistically indistinguishable from FP16 on both closed-loop benchmarks (paired $\Delta = -0.00$, 95% CI $[-0.17, +0.20]$) and is the recipe we certify. Single-scene closed-loop pilots certified three of the failing recipes with perfect scores before full-scale evaluation rejected them. All accuracy conclusions reproduce on a Jetson AGX Orin. We release the harness, the free-drive benchmark, and a practitioner protocol: certify with the full hierarchy, ship INT8, and treat sub-8-bit driving policies as uncertified.

Index Terms—vision-language-action models, autonomous driving, post-training quantization, closed-loop evaluation, safety certification, NeuroNCAP, edge deployment, Jetson Orin

I. Introduction

On-board inference for end-to-end driving stacks motivates compressing the language backbones of driving VLAs, and the prevailing evidence—near-lossless 4-bit weight quantization on open-loop nuScenes planning—

suggests compression is essentially free. This paper shows that the prevailing evidence is wrong in a specific, layered way: it is not only the open-loop metric that is blind, but also the successively stronger closed-loop proxies one would naturally reach for, and the recipes the LLM-quantization literature trusts (group-wise scales, activation-aware scaling, orthogonal rotation) each pass some level of the hierarchy while failing the one that matters.

Contributions:

- A four-level metric hierarchy for certifying quantized driving policies—open-loop L2, generation coherence, route competence, graded closed-loop safety—with an empirical demonstration that each level certifies configurations the next level rejects (Table V).
- A controlled demonstration of open-loop blindness ($19\times$ ego-ablation), and evidence that closed-loop proxies short of graded safety are also insufficient: coherence certifies group-wise INT4 and route competence certifies AWQ-INT4, both of which fail graded safety at naive-INT4 level.
- A free-drive route-competence benchmark on NeuroNCAP scenes (adversary-free, cross-track-gated, anti-freeze-gated, with a collision-artifact taxonomy), which removes the stay-in-place confound that contaminates adversarial scores of degenerate policies.
- A full-scale verdict with confidence intervals: per-channel INT8 is certified (indistinguishable from FP16 on both closed-loop benchmarks, $2\times$ compression), while no 4-bit weight recipe survives; single-scene pilots certified three failing recipes—a caution for closed-loop evaluation practice itself.
- End-to-end on-Orin systems and accuracy measurements showing the conclusions transfer to deployment

TABLE I

Open-loop nuScenes planning context (ST-P3 L2, m). Our contribution is not a new planner but the observation that this metric cannot separate FP16 from quantized policies.

Method	1s	2s	3s	Avg.
UniAD [2]	0.44	0.67	0.96	0.69
VAD [3]	0.17	0.34	0.60	0.37
OpenDriveVLA-7B [1]	0.15	0.31	0.55	0.33
OpenDriveVLA-0.5B (study model)	0.15	0.32	0.57	0.35

hardware.

II. Related Work and Positioning

Driving planners and VLAs. UniAD [2] introduced a planning-oriented, query-based perception stack; VAD [3] vectorized the scene for efficiency. OpenDriveVLA [1] reuses UniAD track+map perception as a vision tower for a Qwen2.5 language/action head that emits trajectories as text, reaching state-of-the-art open-loop L2 (Table I). Open-loop critique. “Is Ego Status All You Need?” [4] showed nuScenes open-loop planning is dominated by ego state; we quantify the effect ($19\times$) as it bears on quantization evaluation, and extend the critique to closed-loop proxies. Closed-loop neural simulation. NeuroNCAP [5] couples a NeuRAD [6] renderer to a safety-test protocol with an impact-speed-graded score; we adopt that graded score as the top of our hierarchy and additionally construct an adversary-free competence benchmark from the same scene reconstructions. LLM weight quantization. RTN, GPTQ [8], and AWQ [7] rely on group-wise scales to bound weight-outlier damage [9], [10]; rotation methods (QuaRot [11]) suppress activation outliers. All are validated on language benchmarks or open-loop accuracy. Our positioning. To our knowledge this is the first study to evaluate driving-VLA quantization across a graded closed-loop safety metric, a closed-loop competence metric, and on-device—and the first to show that the standard recipes pass the cheaper metrics while failing graded safety.

III. Method

A. Model and quantization scope

OpenDriveVLA-0.5B pairs Qwen2.5-0.5B (hidden 896, 24 layers) with UniAD stage-1 track+map perception. We apply post-training quantization to the 168 Linear layers of the Qwen backbone (357.8 M parameters), leaving perception, projectors, embeddings, and the output head in floating point, so observed effects are attributable to backbone precision. Weights are dequantized to the compute dtype for inference (fake-quant): the arithmetic path is unchanged, so accuracy effects are real while packed memory gains are reported separately (Sec. IV-F).

B. Quantized configurations

Six configurations span the standard PTQ design space: FP16 (reference); W8 ($b=8$, per-channel symmetric RTN); naive W4 ($b=4$, per-channel); group W4 ($b=4$, group

size $g=128$); AWQ W4 [7] ($b=4$, $g=128$, activation-aware scales calibrated on a rendered-domain closed-loop rollout); and W4A4+DCT (4-bit weights and activations under an orthonormal DCT rotation $y = (WQ^\top)(Qx)$, QuaRot-style [11]). Group-wise scales cost $16/g = 0.125$ bits/weight at $g=128$ ($\approx 1\%$ footprint).

a) Why granularity should help—and what the argument actually proves.: For a block $\mathcal{B}$ sharing scale $s_{\mathcal{B}} = \max_{j \in \mathcal{B}} |w_j| / (2^{b-1} - 1)$, any weight with $|w_i| < \frac{1}{2}s_{\mathcal{B}}$ rounds to zero; at INT4 a single outlier w^* erases every weight with $w^*/|w_i| > 14$ in its block. Per-channel blocks span the whole row, so one outlier erases row-wide; group-wise confines the damage to g weights. This argument bounds weight-reconstruction error, and it correctly predicts the coherence results (Sec. IV-B). It makes no prediction about closed-loop safety—and empirically (Sec. IV-D) finer granularity does not restore it. We state the mechanism at its true strength: necessary for well-formed generation, not sufficient for safe driving.

C. The four-level evaluation hierarchy

(1) Open-loop L2 (standard): ST-P3 L2 on the 162 nuScenes-mini $\cap$ val samples. The ego-ablation zeroes ego-state and 2s history in the prompt, leaving perception untouched, to test how much of L2 is ego-extrapolation. (2) Generation coherence (closed-loop): fraction of frames with a well-formed (non-degenerate) trajectory during NeuroNCAP rollouts. A degenerate/malformed output triggers a stay-in-place fallback. (3) Route competence (closed-loop, ours): on adversary-free rollouts of the same NeuRAD scene reconstructions (original recorded traffic, no injected threat), a rollout succeeds iff (i) maximum cross-track distance to the logged route stays within $\tau=4$ m (the FP16 on-road envelope; cross-track rather than time-matched error, so a correct path driven at a different speed is not penalized), (ii) driven path length exceeds 5 m (anti-freeze gate: a frozen policy fails rather than scoring “safe”), and (iii) no genuine collision. Collisions are classified by ego-frame geometry at impact: spawn (terminated at frame ≤ 1 ; ego spawned in contact) and rear (struck from behind by logged traffic—the ego lawfully drives slower than the log) are simulation artifacts and excluded; frontal/lateral strikes count. (4) Graded closed-loop safety: the NeuroNCAP score—5 stars if no collision, else $4 - \max(0, 1 - v_{\text{impact}}/v_{\text{ref}})$ —graded by impact speed, so it discriminates between full-speed and mitigated collisions even where the binary collision rate saturates.

D. Closed-loop harness and statistics

A native three-process harness (orchestrator; NeuRAD renderer serving six photorealistic cameras at the driven pose; model node reconstructing the UniAD perception input from rendered frames, with an in-place re-quantization endpoint) runs both benchmarks: 16 adversarial scene/scenario instances (frontal/side/stationary) and 14 free-drive scenes, 10 seeds each per configuration

TABLE II
Open-loop nuScenes-mini ($n=162$), ST-P3 L2 (m).

Setting	L2 (m)	well-formed
FP16	0.33	100%
INT8 (per-channel)	0.33	100%
INT4 (per-channel)	0.33	100%
INT3	1.91	92%
INT2	6.45	0%
FP16, ego-state+history zeroed	6.38 (19 $\times$)	100%

($\approx 2,400$ rollouts across sweeps). All aggregate numbers carry 95% percentile-bootstrap confidence intervals over scenario-level means (10^4 resamples); recipe comparisons use paired per-scenario differences.

E. On-device methodology (capture–replay)

The fused planner input is quantization-independent (perception and embeddings are unquantized), so we capture it on the A100, replay it through the decoder-only planner on a Jetson AGX Orin 64GB at each precision, and score with the same evaluator—isolating device $\times$ quantization without porting UniAD’s custom CUDA ops.

IV. Results

A. Level 1: open-loop L2 is blind

Weight quantization is lossless to INT4 (Table II); INT3/INT2 degradation is generation collapse, not graded planning. The ego-ablation inflates L2 19 $\times$ with perception untouched: the metric is $\approx$ ego-extrapolation and cannot see quantization damage. Every 4-bit configuration in this paper passes this level.

B. Level 2: coherence certifies group-wise INT4—incorrectly

In adversarial closed-loop rollouts, per-channel INT4 collapses generation ($\leq 15\%$ well-formed on 15/16 instances, mean 3.7%) while group-wise INT4 restores well-formedness (mean 73.5% vs FP16 76.0%)—the granularity mechanism of Sec. III-B working as predicted, and a quantization $\times$ domain-shift interaction (the same per-channel INT4 is lossless in-distribution, Table II). By this metric group-wise INT4 appears to retain closed-loop behavior. The next two levels show that conclusion is wrong: well-formed trajectory text is not safe driving.

C. Level 3: route competence certifies AWQ—incorrectly

The free-drive benchmark (Table III) removes the freeze confound: naive W4, whose malformed outputs freeze the ego, drives only 3.1 m on average and fails the progress gate (18.6%)—45% of its collisions are artifacts of being struck while frozen, versus 5% genuine. Group-wise W4 moves and tracks the route but succeeds on only 57.1% (off-route in 21% of rollouts): coherent text, mediocre driving. AWQ-W4, however, drives at near-FP16 competence (87.1%, CIs

TABLE III
Free-drive route competence (clean 7 scenes $\times$ 10 seeds; artifact-corrected; 95% bootstrap CI over scenes). Success = in-corridor (cross-track ≤ 4 m) $\wedge$ moved (≥ 5 m) $\wedge$ no genuine collision.

Config	success	95% CI
FP16	94.3%	[82.9, 100]
W8	97.1%	[91.4, 100]
AWQ W4 g128	87.1%	[74.3, 95.7]
group W4 g128	57.1%	[41.4, 74.3]
W4A4+DCT	38.6%	[10.0, 71.4]
naive W4	18.6%	[0.0, 47.1]

TABLE IV
Graded closed-loop safety: NeuroNCAP score (0–5, impact-speed graded), 16 scene/scenario instances $\times$ 10 seeds, 95% bootstrap CI over instances.

Config	NCAP	95% CI	coll. rate
FP16	4.43	[3.65, 4.93]	12.1%
W8	4.50	[3.82, 4.94]	10.6%
naive W4	2.62	[1.60, 3.63]	60.0%
group W4 g128	2.62	[1.53, 3.70]	53.6%
AWQ W4 g128	2.62	[1.49, 3.71]	51.9%
W4A4+DCT	2.86	[1.87, 3.82]	47.5%

overlapping FP16’s). By this metric AWQ-W4 appears deployable. The final level rejects it.

D. Level 4: graded safety rejects every 4-bit recipe; INT8 is certified

On the graded score (Table IV), every 4-bit recipe collapses to 2.6–2.9—naive, group-wise, AWQ, and rotated alike—losing $\approx 40\%$ of FP16’s safety score and quadrupling the collision rate. The paired group–naive difference is +0.01 with 95% CI $[-0.35, +0.34]$: granularity confers no safety benefit despite fully restoring coherence. AWQ’s near-FP16 route competence coexists with naive-level safety: a policy that drives well and cannot handle emergencies. Per-channel INT8 is statistically indistinguishable from FP16: paired $\Delta = -0.00$, 95% CI $[-0.17, +0.20]$, with matching route competence (Table III). W8 is the recipe this study certifies.

The 4-bit failures are structured: all four recipes fail predominantly on stationary-obstacle scenarios (long-horizon stopping) while passing most frontal ones. The mechanism is open (Sec. VI); the structure suggests a decision-horizon sensitivity rather than a generic accuracy loss.

E. The hierarchy, assembled

Table V is the paper in one view: open-loop certifies everything; coherence catches naive W4 but certifies group-wise; competence catches group-wise but certifies AWQ; only graded safety separates AWQ from W8. No cheaper prefix of the hierarchy reaches the correct verdict.

a) Single-scene pilots also mislead.: Before full-scale evaluation, single-scene closed-loop pilots scored group-W4, AWQ-W4, and W4A4+DCT at a perfect 5.0 NCAP

TABLE V

Each metric certifies configurations the next rejects. ✓ = passes at $\geq 90\%$ of FP16 (within noise for level 1).

Config	open-loop	coherence	competence	safety
naive W4	✓	—	—	—
group W4	✓	✓	—	—
AWQ W4	✓	✓	✓	—
W8	✓	✓	✓	✓

TABLE VI

Measured on AGX Orin (1024-token context $\rightarrow$ 60 tokens, median of 30). Latency is precision-flat under fake-quant; the deployable win is the packed footprint.

Config	median lat.	runtime peak	packed weights
FP16	4.02 s	2.88 GB	0.716 GB
W8 g128	4.01 s	2.88 GB	0.363 GB
INT4 g128	4.02 s	2.88 GB	0.185 GB

(on a frontal scenario each recipe ultimately passes); all three collapsed at 14-scene scale. Closed-loop evaluation does not certify a recipe unless it also spans scenario diversity.

F. On-device (Jetson AGX Orin 64 GB)

Replaying real inputs (Sec. III-E) on the Orin over all 162 mini-val samples reproduces the published FP16 accuracy (0.345 m avg L2, vs 0.35 reported) and matches A100 trajectories to ≈ 1.1 cm (FP16). Notably, the blindness of level 1 reproduces on-device: per-sample W4 plans differ from FP16 by 14–24 cm while aggregate L2 moves < 3 cm—the metric averages the damage away on the deployment hardware itself. The certified W8 recipe packs the backbone to 0.36 GB ($2\times$); the whole planner node fits in 2.88 GB. A packed-INT8 speedup requires a custom kernel and is future work; the edge bottleneck remains UniAD perception throughput, not the quantized planner.

V. Discussion: a certification protocol

(1) Never certify on open-loop metrics—they are ego-extrapolation and pass every configuration that later fails. (2) Do not stop at intermediate closed-loop proxies: well-formedness and route competence each certified a recipe that fails graded safety. (3) Use a graded safety score over diverse scenarios: binary collision saturates; single scenes certified three failing recipes. (4) Gate competence metrics against degenerate behavior: without an anti-freeze gate and collision-artifact taxonomy, a frozen policy scores as safe. (5) Ship INT8; treat sub-8-bit as uncertified for driving until a recipe passes the full hierarchy—granularity, AWQ, and rotation each fix the failure mode one level below the one that matters.

VI. Threats to Validity

Renderer domain gap: rendered inputs are OOD to real nuScenes; claims are relative across configurations under identical rendering. Artifact correction: free-drive

excludes spawn/rear collisions by a geometric classifier; 2 scenes remain unusable (ego spawns in contact) and 1 is ambiguous, so competence CIs are computed on the clean 7 (all-14 numbers, which are consistent, are in the repository). Fake-quant: accuracy effects are real; packed speedups are not yet realized (footprints are measured). One checkpoint, one simulator: a 0.5B model on NeuroNCAP; the stationary-scenario failure of 4-bit recipes is characterized but not yet explained—diagnosing it, and testing whether a mechanism-targeted mixed-precision recipe can pass the hierarchy below 8 bits, is ongoing work. AWQ calibration: one rendered-domain rollout; stronger calibration could shift AWQ’s competence, though its safety failure matches recipes that need no calibration.

VII. Reproducibility

The release includes the open-loop sweeps with ego-ablation, the three-process harness with in-place re-quantization, the adversarial and free-drive scenario generators and runners (run_freedrive.sh, run_positive_hunt.sh), the metric tooling with bootstrap CIs and the collision-artifact classifier (aggregate_ncap_success.py, eval_freedrive_route.py), the on-device suite, and all result JSONs. Every number in this paper is produced by those scripts.

VIII. Conclusion

Certifying a quantized driving VLA is a hierarchy problem: open-loop L2, generation coherence, and route competence each certify recipes that graded closed-loop safety rejects—group-wise INT4 restores syntax, AWQ-INT4 restores competence, and neither restores safety, which every 4-bit recipe fails (2.6–2.9 vs FP16’s 4.4) with failures concentrated on long-horizon stationary scenarios. Per-channel INT8 passes every level of the hierarchy, indistinguishably from FP16, at $2\times$ compression, and reproduces on a Jetson AGX Orin. Until a sub-8-bit recipe passes the full hierarchy, INT8 is the certified deployment point for driving VLAs—and no metric short of graded, scenario-diverse, closed-loop safety testing should be trusted to say otherwise.

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